45. Unique Factorization Domains

1. Yes, 5 is an irreducible in $\mathbb{Z}$.

2. Yes, -17 is an irreducible in $\mathbb{Z}$.

3. No, $14 = 2 \cdot 7$ is not an irreducible in $\mathbb{Z}$.

4. Yes, $2x - 3$ is an irreducible in $\mathbb{Z}[x]$.

5. No, $2x - 10 = 2(x - 5)$ is not an irreducible in $\mathbb{Z}[x]$.

6. Yes, $2x - 3$ is an irreducible in $\mathbb{Q}[x]$.

7. Yes, $2x - 10$ is an irreducible in $\mathbb{Q}[x]$, for 2 is a unit there.

8. Yes, $2x - 10$ is an irreducible in $\mathbb{Z}_{11}[x]$, for 2 is a unit there.

9. (See the answer in the text.)

10. In $\mathbb{Z}[x], 4x^2 - 4x + 8 = (2)(2)(x^2 - x + 2)$. The quadratic polynomial is irreducible because its zeros are complex numbers.

In $\mathbb{Q}[x], 4x^2 - 4x + 8$ is already irreducible because 4 is a unit and the zeros of the polynomial are complex numbers.

In $\mathbb{Z}_{11}[x], 4x^2 - 4x + 8 = (4x + 2)(x + 4)$. We found the factorization by discovering that -4 and 5 are zeros of the polynomial. Note that 2 is a unit.

11. We proceed by factoring the smallest number into irreducibles, and using a calculator, discover which irreducibles divide the larger numbers. We find that $234 = 2 \cdot 117 = 2 \cdot 9 \cdot 13$. Our calculator shows that 9 does not divide 3250, but 2 and 13 do, and both 2 and 13 divide 1690. Thus the gcd's are 26 and -26.

12. We proceed by factoring the smallest number into irreducibles, and using a calculator, discover which irreducibles divide the larger numbers. We find that $448 = 4 \cdot 112 = 4 \cdot 4 \cdot 28 = 2^6 \cdot 7$. Our calculator shows that 7 divides both 784 and 1960, and that the highest power of 2 dividing 784 is 16 while the highest power dividing 1960 is 8. Thus the gcd's are $8 \cdot 7 = 56$ and -56.

13. We proceed by factoring the smallest number into irreducibles, and using a calculator, discover which irreducibles divide the larger numbers. We find that $396 = 6 \cdot 66 = 6 \cdot 6 \cdot 11 = 2^2 \cdot 3^2 \cdot 11$. Our calculator shows that both 11 and 9 divide divides the other 3 numbers, but 2178 and 594 are not divisible by 4, but are divisible by 2. Thus the gcd's are $11 \cdot 9 \cdot 2 = 198$ and -198.

14. $18x^2 - 12x + 48 = 6(3x^2 - 2x + 8)$.

15. Because every nonzero $q \in \mathbb{Q}$ is a unit in $\mathbb{Q}[x]$, we can "factor out" any nonzero rational constant as the (unit) content of this polynomial. For example,

$$(1)(18x^2 - 12x + 48) \quad \text{and} \quad \frac{1}{2}(36x^2 - 24x + 96)$$

are two of an infinite number of possible answers.

16. The factorization is $(1)(2x^2 - 3x + 6)$ because the polynomial is primitive.

17. Because every nonzero $a \in \mathbb{Z}_7$ is a unit in $\mathbb{Z}_7[x]$, we can "factor out" any nonzero constant as the (unit) content of this polynomial. For example,

$$(1)(2x^2 - 3x + 6) \quad \text{and} \quad (5)(6x^2 + 5x + 4)$$

are two of an infinite number of possible answers.

18. The definition is incorrect. Quotients may not exist in D.

Two elements a and b in an integral domain D are **associates** in D if and only if there exists a unit $u \in D$ such that $au = b$.

19. The definition is incorrect. Neither factor can be a unit.

A nonzero element of an integral domain D is an **irreducible** of D if and only if it cannot be factored into a product of two elements of D, neither of which is a unit.

20. The definition is incorrect; there may be no notion of *size* for elements of D.

A nonzero element p of an integral domain D is a **prime** of D if and only if p is not a unit, and p does not divide a product of two elements in D unless p divides one of those two elements.

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22. The irreducibles of $D[x]$ are the irreducibles of D, together with the irreducibles of $F[x]$ which are in $D[x]$ and are furthermore *primitive* polynomials in $D[x]$. (See the paragraph following Lemma 45.26.)

23. The polynomial $2x + 4$ is irreducible in $\mathbb{Q}[x]$ but not in $\mathbb{Z}[x]$.

24. Not every nonzero nonunit of $\mathbb{Z} \times \mathbb{Z}$ has a factorization into irreducibles. For example (1, 0) is not a unit, and every factorization of (1,0) has a factor of the form $(\pm 1, 0)$, which is not irreducible because $(\pm 1, 0) = (\pm 1, 0)(1, 9)$. The only irreducibles of $\mathbb{Z} \times \mathbb{Z}$ are $(\pm 1, p)$ and $(q, \pm 1)$, where p and q are irreducibles in $\mathbb{Z}$.

25. Let p be a prime of D, and suppose that $p = ab$ for some $a, b \in D$. Then $ab = (1)p$, so p divides ab and thus divides either a or b, because p is a prime. Suppose that $a = pc$. Then $p = (1)p = pcb$ and cancellation in the integral domain yields $1 = cb$, so b is a unit of D. Similarly, if p divides b, we conclude that a is a unit in D. Thus either a or b is a unit, so p is an irreducible.

26. Let p be an irreducible in a UFD, and suppose that p divides ab. We must show that either p divides a or p divides b. Let $ab = pc$, and factor ab into irreducibles by first factoring a into irreducibles, then factoring b into irreducibles, and finally taking the product of these two factorizations. Now ab could also be factored into irreducibles by taking p times a factorization of c into irreducibles. Because factorization into irreducibles in a UFD is unique up to order and associates, it must be that an associate of p appears in the first factorization, formed by taking factors of a times factors of b. Thus an associate of p, say up, appears in the factorization of a or in the factorization of b. It follows at once that p divides either a or b.

27. *Reflexive:* $a = a \cdot 1$, so $a \sim a$.

Symmetric: Suppose $a \sim b$, so that $a = bu$ for a unit u. Then u^{-1} is a unit and $b = au^{-1}$, so $b \sim a$.

Transitive: Suppose that $a \sim b$ and $b \sim c$. Then there are units u_1 and u_2 such that $a = bu_1$ and $b = cu_2$. Substituting, we have $a = cu_2u_1 = c(u_2u_1)$. Because the product u_2u_1 of two units is again a unit, we find the $a \sim c$.

28. Let a and b be nonunits in $D^* - U$. Suppose that ab is a unit, so that $(ab)c = 1$ for some $c \in D$. Then $a(bc) = 1$ and a is a unit, contrary to our choice for a. Thus ab is again a nonunit, and $ab \neq 0$ because D has no divisors of zero. Hence $ab \in (D^* - U)$ also.

We see that $D^* - U$ is not a group, for the multiplicative identity is a unit, and hence is not in $D^* - U$.

29. Let $g(x)$ be a nonconstant divisor of the primitive polynomial $f(x)$ in $D[x]$. Suppose that $f(x) = g(x)q(x)$. Because D is a UFD, we know that $D[x]$ is a UFD also. Factor $f(x)$ into irreducibles by factoring each of $g(x)$ and $q(x)$ into irreducibles, and then taking the product of these factorizations. Each nonconstant factor appearing is an irreducible in $D[x]$, and hence is a primitive polynomial. Because the product of primitive polynomials is primitive by Corollary 45.26, we see that the content of $g(x)q(x)$ is the product of the content of $g(x)$ and the content of $q(x)$, and must be the same (up to a unit factor) as the content of $f(x)$. But $f(x)$ has content 1 because it is primitive. Thus $g(x)$ and $q(x)$ both have content 1. Hence $g(x)$ is a product of primitive polynomials, so it is primitive by Corollary 45.26.

30. Let N be an ideal in a PID D. If N is not maximal, then there is a proper ideal N_1 of D such that $N \subset N_1$. If N_1 is not maximal, we find a proper ideal N_2 such that $N_1 \subset N_2$. Continuing this process, we construct a chain $N \subset N_1 \subset N_2 \subset \cdots \subset N_i$ of proper ideals, each properly contained in the next except for the last ideal. Because a PID satisfies the ascending chain condition, we cannot extend this to an infinite such chain, so after some finite number of steps we must encounter a proper ideal N_r that contains N and that is not properly contained in any proper ideal of D. That is, we attain a maximal ideal N_r of D that contains N.

31. We have $x^3 - y^3 = (x-y)(x^2+xy+y^2)$. Of course $x-y$ is irreducible. We claim that x^2+xy+y^2 is irreducible in $\mathbb{Q}[x,y]$. Suppose that x^2+xy+y^2 factors into a product of two polynomials that are not units in $\mathbb{Q}[x,y]$. Such a factorization would have to be of the form $x^2+xy+y^2 = (ax+by)(cx+dy)$ with a, b, c, and d all nonzero elements of $\mathbb{Q}$. Consider the evaluation homomorphism $\phi_1 : (\mathbb{Q}[x])[y] \to \mathbb{Q}[x]$ such that $\phi(y) = 1$. Applying ϕ_1 to both sides of such a factorization would yield $x^2 + x + 1 = (ax+b)(cx+d)$. But x^2+x+1 is irreducible in $\mathbb{Q}[x]$ because its zeros are complex, so no such factorization exists. This shows that x^2+xy+y^2 is irreducible in $(\mathbb{Q}[x])[y]$ which isomorphic to $\mathbb{Q}[x,y]$ under an isomorphism that identifies y^2+yx+x^2 and x^2+xy+y^2.

32. We show that ACC implies MC, that MC implies FBC, and that FBC implies ACC.

ACC *implies* MC*:* Suppose that MC does not hold for some set S of ideals of R; that is, suppose it is not true that S contains an ideal not properly contained in any other ideal of S. Then every ideal of S is properly contained in another ideal of S. We can then start with any ideal N_1 of S and find an ideal N_2 of S properly containing it, then find $N_3 \in S$ properly containing N_2, etc. Thus we could construct an infinite chain of ideals $N_1 \subset N_2 \subset N_3 \subset \cdots$ which contradicts the ACC. Hence the ACC implies the MC.

MC *implies* FBC*:* Suppose the FBC does not hold, and let N be an ideal of R having no finite generating set. Let $b_1 \in N$ and let $N_1 = \langle b_1 \rangle$ be the smallest ideal of N containing b_1. Now $N_1 \neq N$ or $\{b_1\}$ would be a generating set for N, so find $b_2 \in N$ such that $b_2 \notin N_1$. Let N_2 be the intersection of all ideals containing b_1 and b_2. Because N contains b_1 and b_2, we see that $N_2 \subseteq N$, but $N_2 \neq N$ because $\{b_1, b_2\}$ cannot be a generating set for N. We then choose $n_3 \in N$ but not in N_2, and let N_3 be the intersection of all ideals containing b_1, b_2, and b_3. Continuing this process, using the fact that N has no finite generating set, we can construct an infinite chain of ideals $N_1 \subset N_2 \subset N_3 \subset \cdots$ of R. But then the set $S = \{N_i \mid i \in \mathbb{Z}^+\}$ is a set of ideals, each of which is properly contained in another ideal of the set, for $N_i \subset N_{i+1}$. This would contradict the MC, so the FBC is true.

FBC *implies* ACC*:* Let $N_1 \subseteq N_2 \subseteq N_3 \subseteq \cdots$ be a chain of ideals in R, and let $N = \bigcup_{i=1}^{\infty} N_i$. It is easy to see that N is an ideal of R. Let $B_N = \{b_1, b_2, \cdots, b_n\}$ be a finite basis for N. Let $b_j \in N_{i_j}$. If r is the maximm of the subscripts i_j, then $B_N \subseteq N_r$. Because $N_r \subseteq N$ and N is the intersection of all ideals containing B_N, we must have $N_r = N$. Hence $N_r = N_{r+1} = N_{r+2} = \cdots$ so the ACC is satisfied.